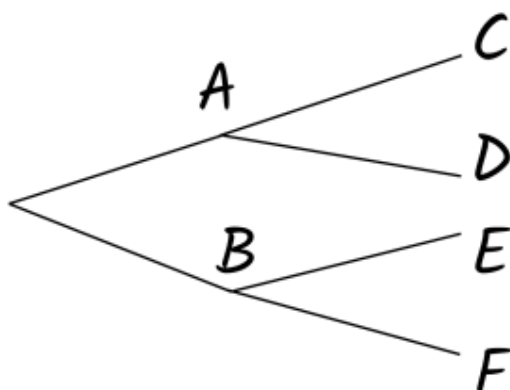


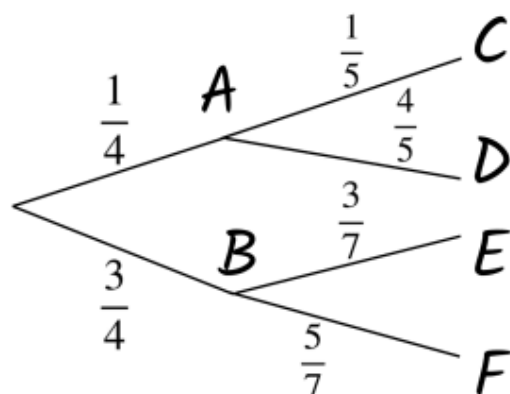
Calculating theoretical probabilities from probability trees (two events)

- 1 If $P(A) = \frac{1}{3}$ and $P(A \text{ and } D) = \frac{1}{5}$, then what is $P(D)$? Tick **1** correct answer



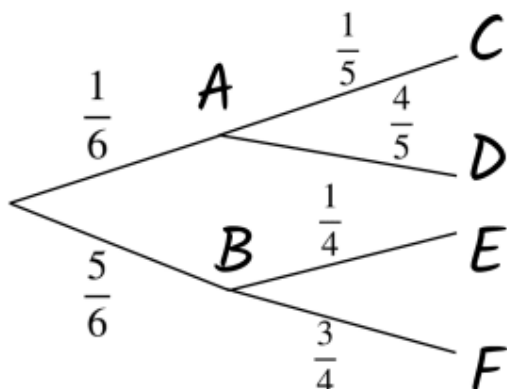
- ☒ $\frac{3}{5}$
☐ $\frac{1}{5}$
☐ $\frac{2}{5}$

- 2 What is the error in this tree diagram? Tick **1** correct answer



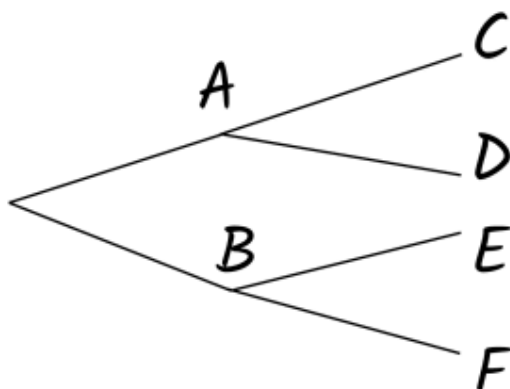
- ☐ Events B and F should sum to 1 but they don't.
☐ Events A and C should sum to 1 but they don't.
☐ Events D and E should sum to 1 but they don't.
☒ Events E and F should sum to 1 but they don't.

- 3 What is a suitable calculation for the probability of event B occurring, and event F not occurring? Tick **1** correct answer



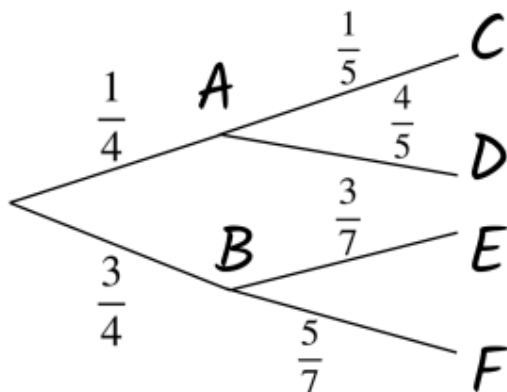
- ☒ $\frac{5}{6} * \frac{1}{4}$
☐ $\frac{5}{6} + \frac{1}{4}$
☐ $\frac{5}{6} * \frac{3}{4}$
☐ $\frac{5}{6} + \frac{3}{4}$

- 4 If the first and second event in this tree diagram were both flipping a coin once, what would be the probability of event B and event F occurring? Tick **1** correct answer



- ☒ $\frac{1}{4}$
☐ $\frac{1}{2}$
☐ $\frac{2}{6}$

- 5 What is the least likely (but not impossible) pair of events occurring in this tree diagram? Tick **1** correct answer



- ☒ A then C
☐ A then D
☐ B then E
☐ B then F

6 Which pair of events has a more than even chance of occurring? Tick **1** correct answer

